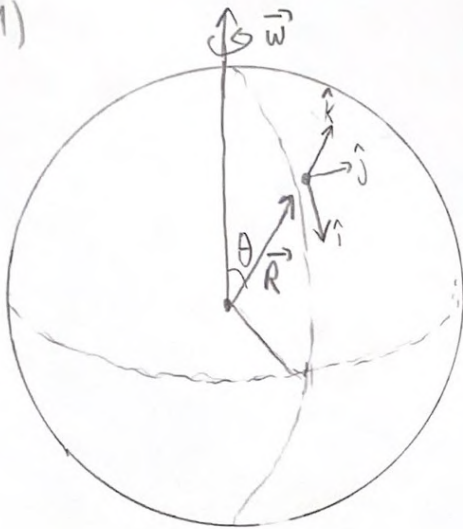
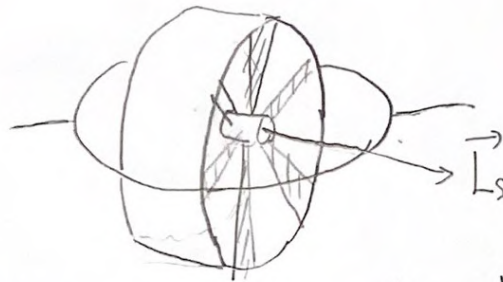


1)



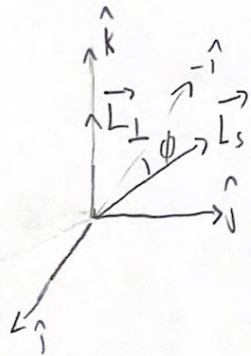
- Choose a ref. frame rotating with $\vec{\omega}$;
- $\hookrightarrow \hat{i}$ is the North direction
- $\hookrightarrow \hat{j}$ is the East direction
- $\hookrightarrow \hat{k}$ is the radial direction



$$\vec{\omega} = \omega \cos \theta \hat{k} - \omega \sin \theta \hat{i}$$

$\rightarrow$ Gyrocompass can spin and rotate in the horizontal plane ($\hat{i}$ - $\hat{j}$ plane)

$\rightarrow$ Say gyrocompass's symmetry axis makes an angle ϕ with $-\hat{j}$ axis: (at any given time)



$\rightarrow$ It's angular momentum will have 2 components.

$\hookrightarrow \vec{L}_s$ due to its spin (along symmetry axis)

$\hookrightarrow \vec{L}_\perp$ due to $\vec{L}_s$'s rotation about $\hat{k}$.

$$\vec{L} = \vec{L}_s + \vec{L}_\perp = I_3 \Omega (\sin \phi \hat{j} - \cos \phi \hat{i}) + I \dot{\phi} \hat{k}$$

where I_3 and I are respectively moment of inertia about symmetry axis and the diameter.

Let's now turn our focus on the torque. Since we're in a rotating ref. frame some fictitious forces will have an effect upon the gyrocompass. One can easily recall from our discussions in class.

$$\vec{F}_{\text{rot}} = m \vec{a}_{\text{rot}} = \vec{F} - \underbrace{2m \vec{\omega} \times \vec{V}_{\text{rot}}}_{\text{Coriolis force}} - \underbrace{m \vec{\omega} \times (\vec{\omega} \times \vec{r})}_{\text{centrifugal force}}$$

Define $\vec{R}$ to be the distance from the center of Earth to the CM of gyrocompass. Also define $\vec{p}_i$ to be the distance from the symmetry axis of gyrocompass to its i^{th} particle. Then

$$\vec{r} = \vec{R} + \vec{p}_i \quad \vec{V}_{\text{rot}} = \vec{\Omega} \times \vec{p}_i$$

$$\vec{F}_{\text{rot}} = \vec{F}_{\text{reel}} - 2 \sum_{i=1}^N m_i \vec{w} \times (\vec{\Omega} \times \vec{p}_i) - \sum_{i=1}^N m_i \vec{w} \times [\vec{w} \times (\vec{R} + \vec{p}_i)]$$

* Now since we're given that $|\vec{\Omega}| \gg |\vec{w}|$, the term $\sum_{i=1}^N m_i \vec{w} \times (\vec{w} \times \vec{p}_i)$ is negligible, so it's safe to disregard it

∇ Note that eliminating the term $\sum_{i=1}^N m_i \vec{w} \times (\vec{w} \times \vec{R})$ would be theoretically faulty since order of $\vec{R}$ can be much larger than $\vec{p}_i$.

$$\vec{\tau} = \underbrace{\vec{p}_i \times \vec{F}_{\text{rot}}}_{\text{w.r.t. CM}} = \sum_{i=1}^N \vec{p}_i \cdot \underbrace{(m_i \vec{g} + \vec{N}_i)}_{\text{real forces}} - 2 \sum_{i=1}^N m_i \vec{p}_i \times [\vec{w} \times (\vec{\Omega} \times \vec{p}_i)] - \sum_{i=1}^N m_i \vec{w} \times (\vec{w} \times \vec{p}_i)$$

* Obviously real forces (gravity and reaction force) will not create any torque on the system since they're constant throughout the motion.

$$\vec{\tau} = -2 \sum_{i=1}^N m_i \vec{p}_i \times [\vec{w} \times (\vec{\Omega} \times \vec{p}_i)] - \underbrace{\vec{w} \times \left(\vec{w} \times \sum_{i=1}^N m_i \vec{p}_i \right)}_{0 \text{ by definition}}$$

Then net torque on the system is found to be:

$$\vec{\tau} = -2 \sum_{i=1}^N m_i \vec{p}_i \times [\vec{w} \times (\vec{\Omega} \times \vec{p}_i)]$$

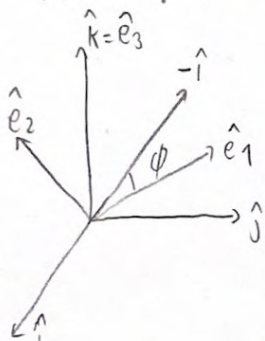
$$\vec{w} = w \cos \theta \hat{k} - w \sin \theta \hat{i}$$

Using the property $A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$

$$\vec{\tau} = -2 \sum_i m_i \vec{p}_i \times [\vec{\Omega} (\vec{w} \cdot \vec{p}_i) - \vec{p}_i (\vec{w} \cdot \vec{\Omega})]$$

$$= -2 \sum_i m_i (\vec{p}_i \times \vec{\Omega}) (\vec{w} \cdot \vec{p}_i) \quad \text{since } \vec{p}_i \times \vec{p}_i = 0.$$

Now let's define a new set of coordinates rotating with gyroscope around $\hat{k}$



Since we are only interested in oscillations around $\hat{e}_3$ axis, we can find the $\hat{e}_3$ component of the torque.

Notice that $\vec{w} = w \cos \theta \hat{e}_3 - w \sin \theta \cos \phi \hat{e}_1 - w \sin \theta \sin \phi \hat{e}_2$

$$\vec{p}_i = p_1 \hat{e}_1 + p_2 \hat{e}_2 + p_3 \hat{e}_3, \quad \vec{\Omega} = \Omega \hat{e}_1$$

$$\vec{\tau} = -2 \sum_i m_i \left[p_1 \Omega (\hat{e}_1 \times \hat{e}_1) + p_2 \Omega (\hat{e}_2 \times \hat{e}_1) + p_3 \Omega (\hat{e}_3 \times \hat{e}_1) \right]$$

$$* [-w \sin \theta \cos \phi p_1 - w \sin \theta \sin \phi p_2 + w \cos \theta p_3]$$

$$\tau_3 = \vec{\tau} \cdot \hat{e}_3 = -2 \sum_i m_i (-p_2 \Omega) [-w \sin \theta \cos \phi p_1 - w \sin \theta \sin \phi p_2 + w \cos \theta p_3]$$

$$= 2 \Omega w \left[-\sin \theta \cos \phi \sum_i m_i p_1 p_2 + \cos \theta \sum_i m_i p_2 p_3 - \sin \theta \sin \phi \sum_i m_i p_2^2 \right]$$

Notice $\sum_{i=1}^N m_i p_1 p_2 = I_{12}$ and $\sum_{i=1}^N m_i p_2 p_3 = I_{23}$ where I is the moment of inertia tensor defined w.r.t. principal axes of the symmetric top. Since I will be diagonal we should have $I_{12} = I_{23} = 0$.

The only term remaining in the expression of τ_3 is:

$$\tau_3 = -\Omega \omega \sin\theta \sin\phi \sum_{i=1}^N 2m_i p_2^2$$

One should notice at this point that $p_2 = p_3$ since top is symmetric around $\hat{e}_1$. Then

$$\sum_{i=1}^N 2m_i p_2^2 = \sum_{i=1}^N m_i (p_2^2 + p_3^2) = \sum_i m_i (p^2 - p_1^2) = I_{11} = I_3 \rightarrow \begin{array}{l} \text{according to} \\ \text{the convention} \\ \text{we've adopted} \\ \text{in class.} \end{array}$$

$$\tau_3 = -\Omega \omega I_3 \sin\theta \sin\phi$$

Also we can write the angular momentum along $\hat{e}_3$ as:

$$\vec{L}_3 = I \dot{\phi}$$

$$\tau_3 = \frac{d\vec{L}_3}{dt} \Rightarrow I \ddot{\phi} = -\Omega \omega I_3 \sin\theta \sin\phi \Rightarrow \ddot{\phi} = -\left(\frac{I_3}{I} \Omega \omega \sin\theta\right) \sin\phi$$

$$\vec{\tau} = -2 \sum_{i=1}^N m_i \vec{r}_i \times \left[w \cos \theta \Omega \hat{p}_i (\hat{k} \times \hat{\phi}') - w \sin \theta \Omega \hat{p}_i (\hat{i} \times \hat{\phi}') \right]$$

$$= -2 \sum_{i=1}^N m_i w \Omega \hat{p}_i^2 \left\{ \cos \theta [\hat{r}'_i \times (\hat{k} \times \hat{\phi}')] - \sin \theta [\hat{r}'_i \times (\hat{i} \times \hat{\phi}')] \right\}$$

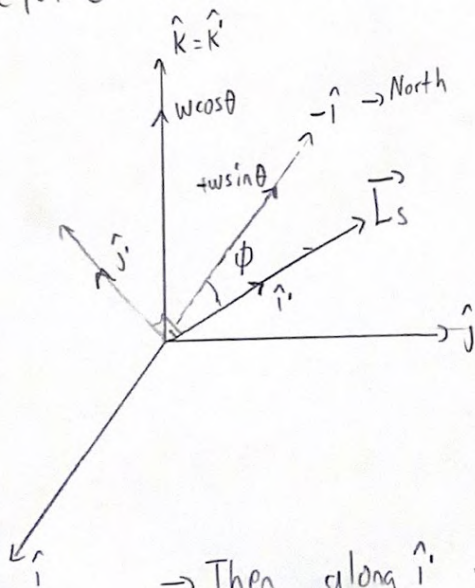
Using the property $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$

$$\hat{r}'_i \times (\hat{k} \times \hat{\phi}') = \hat{k} (\hat{r}'_i \cdot \hat{\phi}') - \hat{\phi}' (\hat{r}'_i \cdot \hat{k})$$

$$\hat{r}'_i \times (\hat{i} \times \hat{\phi}') = \hat{i} (\hat{r}'_i \cdot \hat{\phi}') - \hat{\phi}' (\hat{r}'_i \cdot \hat{i})$$

Alternative Solution Using Lagrangian

In order to avoid from forces and torques one can refer to another approach which doesn't require vector calculations. Namely Lagrangian!



Choose the symmetry axis of gyrocompass as $\hat{i}$ and the perpendicular axis along the diameter (still in $\hat{i}$ - $\hat{j}$ plane) $\hat{j}$ while $\hat{k}'$ remains $\hat{k}$. These describe the orthonormal basis vectors of body of the gyro.

We have also found that $\vec{\omega}$ is projected as

$$\vec{\omega} = w \cos \theta \hat{k} - w \sin \theta \hat{i}$$

→ Then along $\hat{i}$ we have $\underbrace{\Omega}_{\text{spin}} + w \sin \theta \cos \phi$

along $\hat{j}$ we have $w \sin \theta \sin \phi$

along $\hat{k}$ we have $\underbrace{\phi}_{\text{spin}} + w \cos \theta$

due to symmetry axis's rotation about $\hat{k}$

$$\vec{\omega} = \begin{bmatrix} \Omega + w \sin \theta \cos \phi \\ w \sin \theta \sin \phi \\ \dot{\phi} + w \cos \theta \end{bmatrix}$$

$$I = \begin{bmatrix} I_3 & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}$$

→ $L = T - V$ but if one takes the origin as the CM of gyro-compass potential term is basically zero.

$$L = T = \frac{1}{2} I_3 (\Omega + w \sin \theta \cos \phi)^2 + \frac{1}{2} I [w \sin \theta \sin \phi]^2 + (\dot{\phi} + w \cos \theta)^2$$

A feasible question one might ask is the constancy of Ω . To prove this let $\Omega = \dot{\psi}$. Clearly there are no terms that explicitly contain ψ , then. $\frac{\partial L}{\partial \psi} = 0$. But this means $\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\psi}} \right) = 0$

$$\frac{d}{dt} (I_3 \dot{\psi} + 2 w \sin \theta \cos \phi) = 0 \Rightarrow \text{Since } |w| \ll \dot{\psi} = \Omega, \text{ we can ignore the second term on the right}$$

$$\frac{d}{dt} (I_3 \dot{\psi}) \approx 0 \Rightarrow \Omega = \dot{\psi} \text{ is almost constant or variations are negligible}$$

$$L = \frac{1}{2} I_3 [\Omega^2 + 2 \Omega w \sin \theta \cos \phi + w^2 \sin^2 \theta \cos^2 \phi] + \frac{1}{2} I [w^2 \sin^2 \theta \sin^2 \phi + \dot{\phi}^2 + 2 \dot{\phi} w \cos \theta + w^2 \cos^2 \theta]$$

$$\frac{\partial L}{\partial \dot{\phi}} = I \dot{\phi} + I w \cos \theta \quad \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) = I \ddot{\phi} \quad * w \text{ and } \theta \text{ are time independent.}$$

$$\frac{\partial L}{\partial \phi} = -I_3 \Omega w \sin \theta \sin \phi - I_3 w^2 \sin^2 \theta \cos \phi \sin \phi + I w^2 \sin^2 \theta \sin \phi \cos \phi$$

Since we know $|\omega| \ll |\Omega|$ we can ignore the last 2 terms.

$$\frac{\partial L}{\partial \phi} = -I_3 \Omega \omega \sin \theta \sin \phi$$

$$\frac{\partial L}{\partial \phi} = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\phi}} \right) \Rightarrow I \ddot{\phi} = I_3 \Omega \omega \sin \theta \sin \phi$$

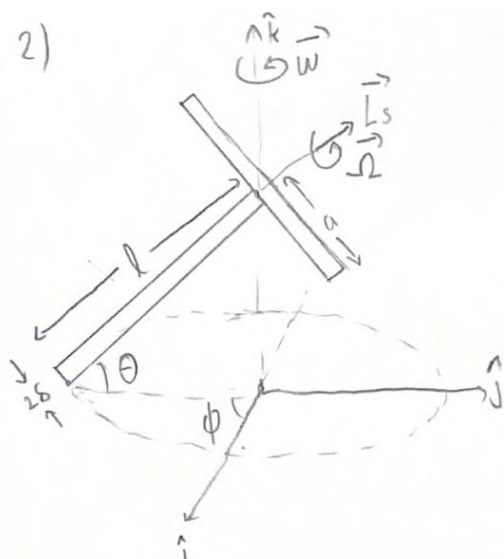
If we release the gyroscope as its symmetry axis is close to North direction so that $\sin \phi \ll 1$.

$$\ddot{\phi} = - \left(\frac{I_3}{I} \Omega \omega \sin \theta \right) \phi \quad \Rightarrow \quad \text{We see that symmetry axis oscillates around north direction harmonically,}$$

with $T = 2\pi \sqrt{\frac{I}{I_3 \Omega \omega \sin \theta}}$

* If there is some kind of damping effect such as friction or air resistance, symmetry axis will point in the North direction after some oscillations.

2)



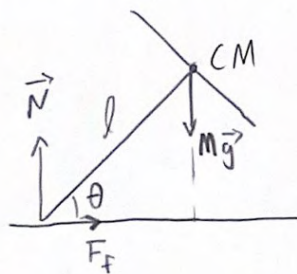
* The tip with radius δ rolls without slipping on a circle whose radius is $l \cos \theta$ (since CM is directly above the center of the circle)

* The top will have a spin angular momentum along $\hat{\Omega}$ and it will precess around $\vec{\omega}$ with $|\vec{\omega}|$.

Rolling without slipping condition $\Rightarrow \Omega \delta = \omega l \cos \theta$

Spin angular momentum L_s at any moment is: $\vec{L}_s = -L_s \cos \phi \hat{i} + L_s \sin \phi \hat{j}$

where $L_s = I \Omega$, Here $I = \frac{1}{2} M a^2$.



* Since CM is directly above the center of the circle there will be no centrifugal acceleration. Then friction force has no radial component ($F_f = 0$) and it only has a tangential component to prevent slipping.

$\vec{N} = M\vec{g}$ required for vertical stability.

$$\vec{C}_{cm} = l \cos \theta N \hat{\phi} = M g l \cos \theta \hat{\phi}$$

Notice that $\vec{\omega}$ and $\dot{\hat{\phi}}$ are related as: $\omega = -\dot{\phi}$

Then total angular momentum can be written as:

$$\vec{L} = L_K \hat{k} + \vec{L}_s \quad \text{but} \quad \frac{d\vec{L}}{dt} = \frac{d\vec{L}_s}{dt} \quad \text{since } L_K \text{ is constant in time}$$

$$\frac{d\vec{L}_S}{dt} = L_S \dot{\phi} \sin\phi \hat{\phi} \hat{i} + L_S \dot{\phi} \cos\phi \hat{\phi} \hat{j} = -L_S \dot{\phi} \hat{\phi}$$

one can inspect the figure to realize
 $\hat{\phi} = \sin\phi \hat{i} + \cos\phi \hat{j}$

$$= L_S \omega \hat{\phi}$$

→ In the inertial frame we're working on we must have:

$$\frac{d\vec{L}_S}{dt} = \tau_{cm} \Rightarrow L_S \omega \hat{\phi} = Mgl \cos\theta \hat{\phi}$$

$$\frac{1}{2} Ma^2 \Omega \omega = Mgl \cos\theta \quad \text{and} \quad \omega = \frac{\Omega \delta}{l \cos\theta} \quad \text{from rolling without slipping condition}$$

$$\frac{1}{2} Ma^2 \Omega^2 \frac{\delta}{l \cos\theta} = Mgl \cos\theta, \quad \cos^2\theta = \frac{\Omega^2 a^2 \delta}{2l^2 g}$$

$$\theta = \cos^{-1} \left[\frac{\Omega a}{l} \sqrt{\frac{\delta}{2g}} \right] \quad \text{or} \quad \Omega = \frac{l \cos\theta}{a} \sqrt{\frac{2g}{\delta}}$$

* For the symmetric top's CM to be on top of circles center we should initially set either Ω or θ values as above.

* As θ gets close to $\pi/2$, Ω decreases as expected. Notice that if we want θ close to 0, we should have a symmetric top with a small radius which inversely increases Ω !